MARY CLAYTON SOTO

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EDUCATION

Duke University, *Pratt School of Engineering* **Aug 2024 – B.S. 2028**

- **Majors:** Biomedical Engineering & Mathematics; **Minor:** Visual Arts
- **GPA:** 3.92/4.00
- **Activities & Societies:** Pratt Tour Guide, Club Soccer, Theta Tau Engineering Fraternity, Phi Alpha Delta, Tri Delta
- **Relevant Coursework:** Quantitative Physiology with Biostatistical Applications, Engineering Design and Collaboration, Mechanics of Solids, Materials Science, Signals and Systems, Modeling Cellular and Molecular Systems

Maclay School, *Salutatorian* **May 2024**

ENGINEERING EXPERIENCE

Duke Surgical Education & Assessment Lab, *Research Assistant* **Mar 2025 – Present**

- Design and iterate hydrogel organ models across 3 organ types, segmenting MRI-derived DICOM scans in 3D Slicer and designing and 3D printing Fusion 360 molds, tuning materials to replicate human tissue mechanics for surgical simulation.
- Help develop a dissolvable core molding process that casts CT-derived gallbladders around a dissolvable PVA core, producing hollow, fluid-bearing models for laparoscopic training.
- Contribute to a computer-vision system that scores surgical knot-tying from video, adapting video intake to handle recording variability across 1,000+ recordings.

FSU Microparticle Lab, *Research Assistant* **May – Aug 2025**

- Developed a MATLAB pipeline to segment and classify fluorescence cell images, analyzing ~20,000 cells and generating a standardized dataset for peer-reviewed publication.
- Automated cell quantification through optimized convolution-based segmentation and thresholding, improving classification accuracy across heterogeneous fluorescence channels and saving hundreds of hours of manual counting.

Tallahassee Research Institute, *Research Coordinator* **May – Aug 2026**

- Managed clinical research workflows across 10+ cardiovascular studies, coordinating protocols and documentation.
- Compiled and analyzed patient data into structured reports to support ongoing device trials and investigator decision-making.

ACADEMIC PROJECTS

Rollator Design Project **Aug 2024 – May 2025**

- Designed a wheeled-cage oxygen canister mount for Duke Hospital rollators through three prototype rounds, ending in a SolidWorks-modeled, waterjet-cut aluminum build that let post-surgery patients walk independently and freed a nurse.
- Verified the design against 7 criteria, reducing tipping angle by 6° and earning a 4.3/5 usability rating from 30 evaluators.

Biomedical Image Segmentation Pipeline **Jan – May 2026**

- Built and compared three Python pipelines for cell image segmentation (Fourier analysis, convolution-based edge detection, thresholding), finding convolution alone captured most boundaries while frequency enhancement improved low-contrast detail.

Computational Modeling of Engineered Probiotic Therapeutics **Jan – May 2026**

- Modeled an engineered probiotic yeast that degrades gut PET microplastics, coupling a TPA-responsive gene circuit, Michaelis-Menten kinetics, and gut transport; compared three PETase variants, predicting up to ~24% degradation in two weeks.

SuperComp: Personal Training Readiness Pipeline **May – Aug 2026**

- Built a Python pipeline merging ~1.2M heart-rate samples and 3,151 workouts from Oura, Peloton, and Apple Health into a daily readiness score, then backtested it on 386 days and reframed it as a guardrail when it proved non-predictive.

LEADERSHIP & INVOLVEMENT

EGR105: Coding for Engineers, *Teaching Assistant* **Aug 2025 – May 2026**

- Mentored 50+ first-year engineering students in Python algorithm design, reinforcing debugging workflows and Git version control.
- Taught NumPy and Matplotlib for data analysis and visualization, emphasizing efficient data handling and interpretation.

Theta Tau, *Vice President of Professional Development* **Aug 2025 – Present**

- Execute events to highlight engineering research and innovation, expanding 100+ members' professional opportunities and skills.
- Created a centralized member resource database for academic planning, networking, and career development.

PUBLICATIONS & PRESENTATIONS

- Fukuda, M.; **Soto, M. C.**; et al. "Engineered Microparticles Suggest Proteolysis as a Critical Prerequisite for Chromatin Clearance in Macrophage Phagosomes." ACS Appl. Mater. Interfaces, 2026, 18, 16137–16147.
- Gao, S.; Yamamoto, K. K.; **Soto, M. C.**; et al. "Dissolvable Core Molding (DCM): A Gallbladder Model Toward Fluid-Bearing Organ Simulators." ACS Surgical Simulation Summit, 2027 (accepted).

TECHNICAL SKILLS

- **Languages & Software:** Python (NumPy, pandas, SciPy, OpenCV), MATLAB, R, C++, Git, SolidWorks, Fusion 360, 3D Slicer
- **Fabrication:** 3D printing (FDM, resin), laser & waterjet cutting, machining, silicone & hydrogel molding and casting
- **Analysis & Modeling:** statistical analysis (ANOVA, regression, hypothesis testing), ODE modeling, image processing